

Skills Summary

Multistep Equations with Rational Coefficients

Use this reference while completing your worksheet or when studying.

Skill 1 — Clearing the fractions first

Rule: multiply *both sides* of the equation by a common denominator. Every term gets multiplied — including any term that had no fraction in it. After that step the equation contains no fractions at all and is an ordinary two-step equation. The common denominator is the smallest number all the denominators divide into.

Example: Solve $\frac{x}{2} + \frac{x}{3} = 10$.

Step 1. Two different denominators make the arithmetic messy, so I clear them both at once rather than combining the fractions.

Step 2. Denominators 2 and 3 divide into 6; multiplying every term by 6 gives $3x + 2x = 60$.

Step 3. Collect and divide: $5x = 60$, so $x = 12$. Check: $6 + 4 = 10$.

$$\Rightarrow x = 12$$

Try it: Solve $\frac{x}{3} + \frac{x}{6} = 4$.

check: $8 = x$

Skill 2 — A fraction in front of a bracket

Two legal routes, same answer: undo the fraction first — multiply both sides by the reciprocal, so $\frac{a}{b}(\cdot) = c$ becomes $(\cdot) = \frac{bc}{a}$ — or distribute the fraction across both terms in the bracket and then solve. Pick the first when the right-hand side divides cleanly; pick the second when it does not.

Example: Solve $\frac{2(x-4)}{3} = 6$.

Step 1. The whole bracket is multiplied by a fraction, and the right-hand side divides cleanly, so undoing the fraction is quicker than distributing.

Step 2. Multiply both sides by 3: $2(x-4) = 18$. Divide both sides by 2: $x-4 = 9$.

Step 3. Add 4 to both sides. Check: $2(13-4) \div 3 = 18 \div 3 = 6$.

$$\Rightarrow x = 13$$

Try it: Solve $\frac{3(x+2)}{4} = 6$.

check: $9 = x$

Skill 3 — The variable on both sides

Order of work: clear the fractions, then move every variable term to one side with a both-sides addition or subtraction, then move the numbers to the other side, then divide. If the variable terms cancel and leave a *true* statement such as $5 = 5$, every number works; if they leave a *false* one such as $-2 = 4$, nothing works and there is no solution.

Example: Solve $\frac{3x}{5} - 2 = \frac{x}{10} + 3$.

Step 1. There is an x on each side and two denominators, so I clear the fractions before trying to gather the variable.

Step 2. Multiply every term by 10: $6x - 20 = x + 30$.

Step 3. Subtract x from both sides: $5x - 20 = 30$. Add 20: $5x = 50$. Divide by 5.

$$\Rightarrow x = 10$$

Try it: Solve $\frac{x}{2} + 3 = \frac{x}{6} + 7$.

check: $x = 12$